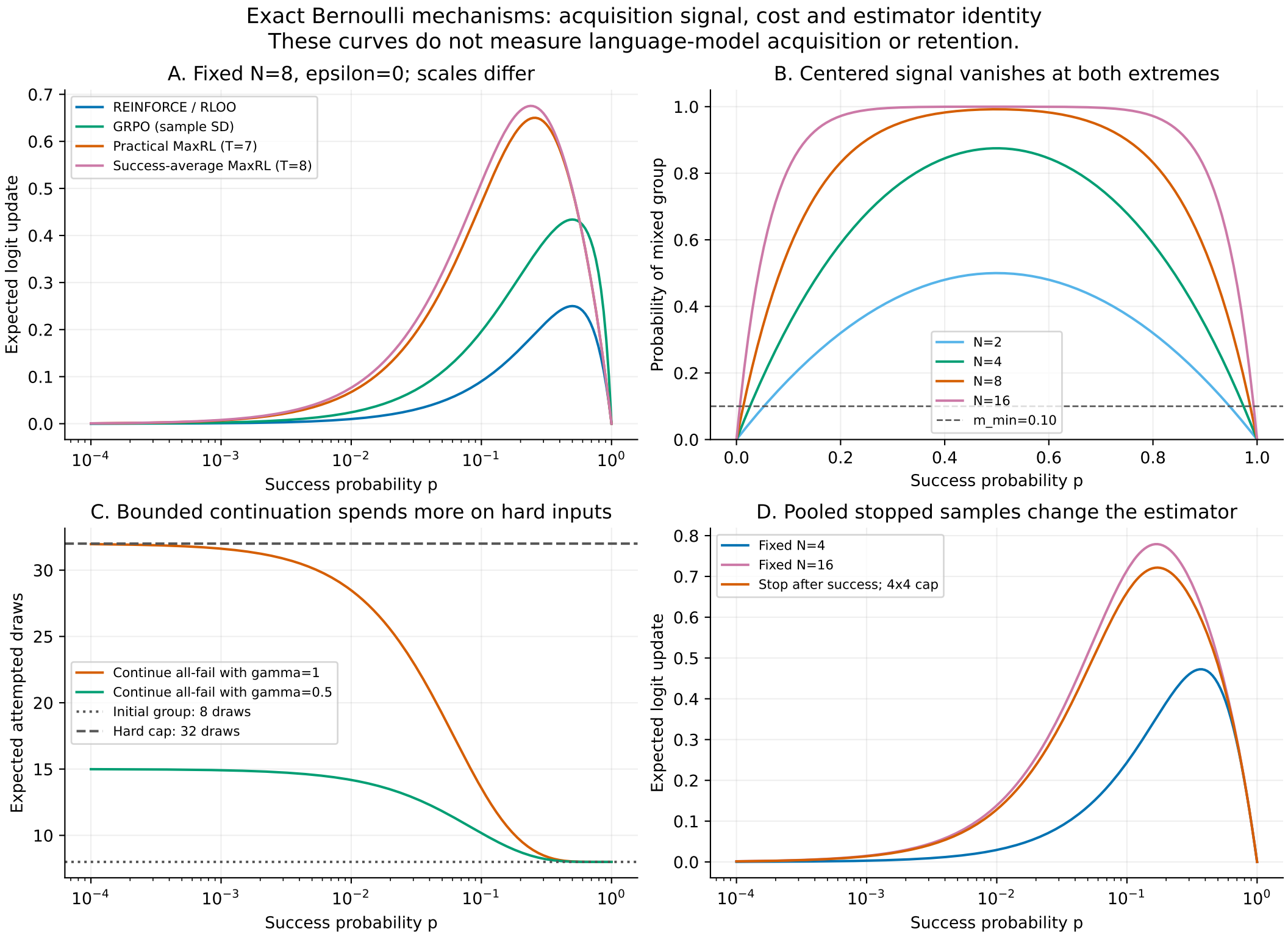
Exact Bernoulli mechanisms: acquisition signal, cost and estimator identity
These curves do not measure language-model acquisition or retention.
A. Fixed N=8, epsilon=0; scales differ
REINFORCE / RLOO
GRPO (sample SD)
Practical MaxRL (T=7)
Success-average MaxRL (T=8)
Expected logit update
Success probability p
B. Centered signal vanishes at both extremes
N=2
N=4
N=8
N=16
m_min=0.10
Probability of mixed group
Success probability p
C. Bounded continuation spends more on hard inputs
Continue all-fail with gamma=1
Continue all-fail with gamma=0.5
Initial group: 8 draws
Hard cap: 32 draws
Expected attempted draws
Success probability p
D. Pooled stopped samples change the estimator
Fixed N=4
Fixed N=16
Stop after success; 4x4 cap
Expected logit update
Success probability p